

Air System Heat Balance Summary for HP-101-FITNESS

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	21897	12470	-	6686	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	880 CFM	0	-	880 CFM	0	-
Ventilation Load	335 CFM	10721	-8240	335 CFM	16850	0
Supply Fan Load	1215 CFM	1801	-	1215 CFM	-1801	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	34419	4230	-	21735	0
Central Cooling Coil	-	34507	4400	-	0	0
Central Heating Coil	-	0	-	-	21807	-
>> Total Conditioning	-	34507	4400	-	21807	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	198 sqft	697	-	198 sqft	438	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	116 sqft	965	-	116 sqft	985	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1280 sqft	2871	-	1280 sqft	2280	-
Interior Wall Convection	1752 sqft	3364	-	1752 sqft	663	-
Ceiling Convection	1280 sqft	4412	-	1280 sqft	1227	-
Overhead Lighting Convection	737 W	1116	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	1928 W	4933	-	0 W	0	-
People Convection	13	2524	12918	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	20883	12918	-	5592	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-102-OFFICES

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	5963	1029	-	3217	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	333 CFM	0	-	333 CFM	0	-
Ventilation Load	73 CFM	2325	-1025	73 CFM	3656	0
Supply Fan Load	333 CFM	493	-	333 CFM	-493	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	8782	4	-	6380	0
Central Cooling Coil	-	8806	0	-	0	0
Central Heating Coil	-	0	-	-	6396	-
>> Total Conditioning	-	8806	0	-	6396	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	281 sqft	579	-	281 sqft	723	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	774 sqft	564	-	774 sqft	998	-
Interior Wall Convection	2203 sqft	1211	-	2203 sqft	440	-
Ceiling Convection	774 sqft	1153	-	774 sqft	408	-
Overhead Lighting Convection	330 W	499	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	319 W	816	-	0 W	0	-
People Convection	5	387	1079	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	5209	1079	-	2569	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-103-BOH EMPLOYEE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - SEPTEMBER 15:00			DESIGN HEATING		
	OA DB / WB 101.0 F / 63.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	10253	1114	-	4132	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	509 CFM	0	-	509 CFM	0	-
Ventilation Load	80 CFM	2176	-1104	80 CFM	4041	0
Supply Fan Load	584 CFM	866	-	584 CFM	-866	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	13296	10	-	7307	0
Central Cooling Coil	-	13314	0	-	0	0
Central Heating Coil	-	0	-	-	7324	-
>> Total Conditioning	-	13314	0	-	7324	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - SEPTEMBER 15:00			DESIGN HEATING		
	OA DB / WB 101.0 F / 63.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	264 sqft	1434	-	264 sqft	711	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	58 sqft	956	-	58 sqft	510	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	594 sqft	1108	-	594 sqft	1519	-
Interior Wall Convection	1564 sqft	1869	-	1564 sqft	539	-
Ceiling Convection	594 sqft	1644	-	594 sqft	599	-
Overhead Lighting Convection	221 W	335	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	778 W	1992	-	0 W	0	-
People Convection	10	671	1167	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	10008	1167	-	3879	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-104-BOH LAUNDRY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	22546	4314	-	6734	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	1293 CFM	0	-	1293 CFM	0	-
Ventilation Load	188 CFM	6033	-3441	188 CFM	9443	0
Supply Fan Load	1293 CFM	1916	-	1293 CFM	-1916	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	30496	873	-	14260	0
Central Cooling Coil	-	30555	881	-	0	0
Central Heating Coil	-	0	-	-	14301	-
>> Total Conditioning	-	30555	881	-	14301	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	680 sqft	2658	-	680 sqft	1630	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1318 sqft	1963	-	1318 sqft	2883	-
Interior Wall Convection	2551 sqft	2813	-	2551 sqft	595	-
Ceiling Convection	1318 sqft	3118	-	1318 sqft	928	-
Overhead Lighting Convection	467 W	707	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	3793 W	9706	-	0 W	0	-
People Convection	10	876	4502	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	21842	4502	-	6035	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-105-LOBBY/MARKET

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	18047	5939	-	9701	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	918 CFM	0	-	918 CFM	0	-
Ventilation Load	298 CFM	9525	-5485	298 CFM	14973	0
Supply Fan Load	918 CFM	1360	-	918 CFM	-1360	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	28933	454	-	23313	0
Central Cooling Coil	-	29020	586	-	0	0
Central Heating Coil	-	0	-	-	23377	-
>> Total Conditioning	-	29020	586	-	23377	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	508 sqft	1939	-	508 sqft	1346	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	116 sqft	1396	-	116 sqft	1001	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1302 sqft	3254	-	1302 sqft	3704	-
Interior Wall Convection	497 sqft	915	-	497 sqft	300	-
Ceiling Convection	1302 sqft	4484	-	1302 sqft	1649	-
Overhead Lighting Convection	448 W	679	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	325 W	833	-	0 W	0	-
People Convection	29	2154	6008	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	15654	6008	-	8000	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	16379	5046	-	9241	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	833 CFM	0	-	833 CFM	0	-
Ventilation Load	524 CFM	16762	-5046	524 CFM	26348	0
Supply Fan Load	833 CFM	1234	-	833 CFM	-1234	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	34375	0	-	34356	0
Central Cooling Coil	-	34566	0	-	0	0
Central Heating Coil	-	0	-	-	34468	-
>> Total Conditioning	-	34566	0	-	34468	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	140 sqft	498	-	140 sqft	369	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	329 sqft	2649	-	329 sqft	2882	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1092 sqft	2653	-	1092 sqft	3133	-
Interior Wall Convection	0 sqft	0	-	0 sqft	0	-
Ceiling Convection	1092 sqft	3848	-	1092 sqft	1529	-
Overhead Lighting Convection	317 W	480	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	546 W	1398	-	0 W	0	-
People Convection	44	3015	5243	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	14540	5243	-	7913	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-107-BUFFET/SERVING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	14929	2588	-	4763	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	555 CFM	0	-	555 CFM	0	-
Ventilation Load	224 CFM	7164	-2588	224 CFM	11262	0
Supply Fan Load	780 CFM	1156	-	780 CFM	-1156	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	23249	0	-	14870	0
Central Cooling Coil	-	23328	0	-	0	0
Central Heating Coil	-	0	-	-	14918	-
>> Total Conditioning	-	23328	0	-	14918	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	275 sqft	1035	-	275 sqft	668	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	747 sqft	1839	-	747 sqft	1837	-
Interior Wall Convection	273 sqft	577	-	273 sqft	136	-
Ceiling Convection	747 sqft	2686	-	747 sqft	850	-
Overhead Lighting Convection	550 W	833	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	2241 W	5735	-	0 W	0	-
People Convection	10	837	2689	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	13541	2689	-	3491	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-108-FOOD PREP

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	19204	3615	-	3622	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	593 CFM	0	-	593 CFM	0	-
Ventilation Load	487 CFM	15567	-3615	487 CFM	24470	0
Supply Fan Load	1080 CFM	1601	-	1080 CFM	-1601	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	36372	0	-	26491	0
Central Cooling Coil	-	36561	0	-	0	0
Central Heating Coil	-	0	-	-	26595	-
>> Total Conditioning	-	36561	0	-	26595	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	314 sqft	1360	-	314 sqft	677	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	696 sqft	1906	-	696 sqft	1633	-
Interior Wall Convection	741 sqft	1824	-	741 sqft	228	-
Ceiling Convection	696 sqft	2810	-	696 sqft	569	-
Overhead Lighting Convection	512 W	775	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	3478 W	8900	-	0 W	0	-
People Convection	14	1169	3756	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	18744	3756	-	3106	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-109-DINING FLEX

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Table 1. System Loads

	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	6662	1675	-	3158	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	361 CFM	0	-	361 CFM	0	-
Ventilation Load	174 CFM	5591	-1675	174 CFM	8750	0
Supply Fan Load	361 CFM	536	-	361 CFM	-536	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	12789	0	-	11373	0
Central Cooling Coil	-	12854	0	-	0	0
Central Heating Coil	-	0	-	-	11410	-
>> Total Conditioning	-	12854	0	-	11410	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	92 sqft	446	-	92 sqft	238	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	58 sqft	500	-	58 sqft	484	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	363 sqft	942	-	363 sqft	1014	-
Interior Wall Convection	485 sqft	997	-	485 sqft	294	-
Ceiling Convection	363 sqft	1296	-	363 sqft	501	-
Overhead Lighting Convection	105 W	160	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	181 W	464	-	0 W	0	-
People Convection	15	1001	1741	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	5806	1741	-	2530	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-110-DINING SMALL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	12727	3291	-	6175	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	703 CFM	0	-	703 CFM	0	-
Ventilation Load	342 CFM	10618	-3287	342 CFM	17199	0
Supply Fan Load	703 CFM	1042	-	703 CFM	-1042	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	24387	3	-	22332	0
Central Cooling Coil	-	24509	0	-	0	0
Central Heating Coil	-	0	-	-	22406	-
>> Total Conditioning	-	24509	0	-	22406	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	98 sqft	449	-	98 sqft	257	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	196 sqft	2717	-	196 sqft	1662	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	713 sqft	2053	-	713 sqft	1998	-
Interior Wall Convection	242 sqft	507	-	242 sqft	147	-
Ceiling Convection	713 sqft	2559	-	713 sqft	990	-
Overhead Lighting Convection	207 W	314	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	356 W	912	-	0 W	0	-
People Convection	29	1968	3422	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	11479	3422	-	5053	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-111-MEETING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Table 1. System Loads

	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	21796	5216	-	8118	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	1281 CFM	0	-	1281 CFM	0	-
Ventilation Load	180 CFM	4685	-4334	180 CFM	9062	0
Supply Fan Load	1281 CFM	1899	-	1281 CFM	-1899	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	28379	882	-	15281	0
Central Cooling Coil	-	28404	953	-	0	0
Central Heating Coil	-	0	-	-	15320	-
>> Total Conditioning	-	28404	953	-	15320	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	244 sqft	1436	-	244 sqft	694	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	184 sqft	3634	-	184 sqft	1579	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	809 sqft	3127	-	809 sqft	2649	-
Interior Wall Convection	1106 sqft	4421	-	1106 sqft	842	-
Ceiling Convection	809 sqft	4891	-	809 sqft	1306	-
Overhead Lighting Convection	483 W	732	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	329 W	843	-	0 W	0	-
People Convection	26	1937	5403	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	21021	5403	-	7070	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-112-ENGINEERING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	1541	365	-	464	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	77 CFM	0	-	77 CFM	0	-
Ventilation Load	23 CFM	743	-365	23 CFM	1169	0
Supply Fan Load	77 CFM	115	-	77 CFM	-115	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	2399	1	-	1518	0
Central Cooling Coil	-	2406	0	-	0	0
Central Heating Coil	-	0	-	-	1523	-
>> Total Conditioning	-	2406	0	-	1523	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	0 sqft	0	-	0 sqft	0	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	233 sqft	142	-	233 sqft	45	-
Interior Wall Convection	688 sqft	304	-	688 sqft	276	-
Ceiling Convection	233 sqft	317	-	233 sqft	114	-
Overhead Lighting Convection	96 W	145	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	116 W	298	-	0 W	0	-
People Convection	2	137	381	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	1342	381	-	435	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-STAIRS1

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Table 1. System Loads

	DESIGN COOLING - JULY 13:00			DESIGN HEATING		
	OA DB / WB 102.9 F / 65.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	10963	0	-	9174	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	512 CFM	0	-	512 CFM	0	-
Ventilation Load	0 CFM	0	0	0 CFM	0	0
Supply Fan Load	512 CFM	758	-	512 CFM	-758	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	11721	0	-	8416	0
Central Cooling Coil	-	11721	0	-	0	0
Central Heating Coil	-	0	-	-	8416	-
>> Total Conditioning	-	11721	0	-	8416	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 13:00			DESIGN HEATING		
	OA DB / WB 102.9 F / 65.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	1263 sqft	3618	-	1263 sqft	3274	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	20 sqft	92	-	20 sqft	128	-
Floor Convection	993 sqft	1207	-	993 sqft	1728	-
Interior Wall Convection	1296 sqft	972	-	1296 sqft	709	-
Ceiling Convection	995 sqft	1887	-	995 sqft	1439	-
Overhead Lighting Convection	328 W	497	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	0 W	0	-	0 W	0	-
People Convection	0	0	0	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	8273	0	-	7278	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Heat Balance Summary for HP-STAIRS2

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	13388	0	-	9452	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	607 CFM	0	-	607 CFM	0	-
Ventilation Load	0 CFM	0	0	0 CFM	0	0
Supply Fan Load	607 CFM	900	-	607 CFM	-900	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	14288	0	-	8553	0
Central Cooling Coil	-	14288	0	-	0	0
Central Heating Coil	-	0	-	-	8553	-
>> Total Conditioning	-	14288	0	-	8553	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	1269 sqft	4916	-	1269 sqft	3303	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	20 sqft	161	-	20 sqft	130	-
Floor Convection	1001 sqft	1354	-	1001 sqft	1757	-
Interior Wall Convection	1293 sqft	1168	-	1293 sqft	726	-
Ceiling Convection	1002 sqft	2242	-	1002 sqft	1461	-
Overhead Lighting Convection	331 W	501	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	0 W	0	-	0 W	0	-
People Convection	0	0	0	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	-	10343	0	-	7377	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.